

Model Assessment and Selection for Prediction - Part 1

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Model Complexity

Generalization
Performance

Model Assessment
and Selection

Assessing EPE

Estimation of
In-Sample Error

Ex: King County Birth
Weight Data

Best subsets regression

Cross-validation

Model complexity

- ▶ In regression analyses, we can base model selection on a pre-specified set of predictor variables
 - ▶ variable selection which includes/excludes a particular variable ('best' subsets regression)
 - ▶ shrinkage methods which include all predictors but controls the size of the coefficients
- ▶ Each approach employs a measure of 'complexity'
 - ▶ number of covariates
 - ▶ amount of control on the size of a coefficient
- ▶ Generically we will refer to this measure as a *tuning parameter*

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- ▶ Determining a specific value for the tuning parameter is part of the model selection process
- ▶ For best subsets regression the tuning parameter is fairly easy to conceptualize, mainly because we can think in terms of the interpretation of predictors and their associated coefficients
- ▶ Other classes of restricted estimators also have associated measures of complexity

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- ▶ Again, in each case we can still embed the choice of tuning parameter into the model selection process
 - ▶ in particular, we can view the determination of the level of complexity of our model as a model selection problem
- ▶ The selection process requires a means of assessing any given model
 - ▶ test or generalization error
 - ▶ error observed in an independent sample
- ▶ Our goal is to develop tools for the joint tasks of model assessment and selection

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Generalization performance

- ▶ We can formalize model assessment via a loss function and use expected prediction error, EPE, as a criterion for choosing a model
 - ▶ choose $f(\cdot)$ which minimizes EPE

$$f^*(\cdot) = \operatorname{argmin}_{f(\cdot)} E[L(Y, f(X))]$$

- ▶ Two examples of commonly considered loss functions are
 1. Squared error (L_2) loss: $E(Y - f(X))^2$
 2. Absolute (L_1) loss: $E|Y - f(X)|$

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- ▶ L_2 loss is commonly used for many reasons, and in this case the we have $f^*(\cdot) = E[Y | X = x]$, the conditional expectation or regression function
- ▶ In this case there are many ways we can estimate $E[Y | X = x]$, and we would like a framework that can be used to assess, and order, competing choices.

Generalization performance

Generalization performance

- ▶ For a specified outcome variable Y and vector of predictor variables X , suppose we have a prediction model $\hat{f}(X)$, the form of which has been determined on the basis of a *training sample*
- ▶ We measure errors between Y and $\hat{f}(X)$ by specifying a loss function $L(Y, \hat{f}(X))$
- ▶ The *test* or *generalization* error is the expected prediction error over an *independent* test sample

$$\text{EPE} = E_{X,Y} [L(Y, \hat{f}(X))]$$

- ▶ the expectation is taken over the joint distribution of X and Y
- ▶ the average error, were the prediction model to be applied to an independent sample from the population

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Generalization performance

- ▶ If we knew the true joint distribution of (X, Y) , we could evaluate this expression directly
 - ▶ feasible in a simulation study where we know the truth
- ▶ However, in real life situations we won't know this joint distribution and so, for a given $\hat{f}(X)$, we need to estimate EPE
- ▶ A tempting choice could be the *training error*

$$\text{err} = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(x_i))$$

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Generalization performance

- ▶ Unfortunately the training error is not a good estimate of test error
 - ▶ the problem is that the estimate $\hat{y}_i = \hat{f}(x_i)$ uses y_i
 - ▶ the solution is specifically chosen because it does well in predicting the training data
- ▶ More specifically, the training error consistently decreases with model complexity
 - ▶ an extreme case is including a parameter for every observation (a *saturated* model), so that $\hat{f}(x_i) = y_i$ and there is zero training error!
- ▶ A model with zero training error can be viewed as an overfit to the training data and will typically generalize poorly
 - ▶ high sampling variability

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Model assessment and selection

- ▶ We've already identified two separate goals we might have in mind: model selection and model assessment
- ▶ Model selection deals with estimating the performance of competing models in order to choose the best one
 - ▶ estimate the test error distribution across these models
 - ▶ choose the model which corresponds to the minimum
- ▶ Model assessment deals with evaluating the generalization error when applying the final model to new data
 - ▶ the final model is still chosen on the basis of the training data
 - ▶ seek an honest assessment of generalization error

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Model assessment and selection

Model assessment and selection

- ▶ In a data-rich situation, we could approach these goals jointly by splitting the data into three parts:

$$\begin{bmatrix} \text{Training} \\ \text{data} \end{bmatrix} \begin{bmatrix} \text{Validation} \\ \text{data} \end{bmatrix} \begin{bmatrix} \text{Test} \\ \text{data} \end{bmatrix}$$

- ▶ Training data: *fit the models*
 - ▶ obtain point estimates for any given model under consideration
 - ▶ repeated use across models
- ▶ Validation data: *choose between models*
 - ▶ estimate the prediction error for model selection
 - ▶ repeated use across models
- ▶ Test data: *estimate generalization error of the final model*
 - ▶ one-time use, at the end of the analysis

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Model assessment and selection

- ▶ Typically, we are not in a position to split the data into three parts
- ▶ A compromise might be to split the data into two parts

$$\begin{bmatrix} \text{Training} \\ \text{data} \end{bmatrix} \quad \begin{bmatrix} \text{Test} \\ \text{data} \end{bmatrix}$$

and approximate the validation step

- ▶ analytically: C_p , AIC and BIC
 - ▶ efficiency sample re-use: cross-validation and the bootstrap
- ▶ Even still, it may not be that splitting into two parts is feasible
- ▶ consider whether or not these methods can be used to obtain reasonable assessments of generalization error

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Assessing EPE

- ▶ Earlier, we noted that the training err

$$\text{err} = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(x_i))$$

would not typically be a good estimate of EPE

- ▶ In particular, we would expect err to be somewhat lower than the true EPE
 - ▶ that is, the estimate would be overly optimistic
- ▶ Part of the discrepancy is due to where the evaluation points occur
 - ▶ EPE refers to expected error on an independent sample
 - ▶ referred to as *extra-sample* error

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- ▶ Methods that directly estimate the extra-sample error include cross-validation and the bootstrap
 - ▶ both involve the clever use and re-use of the training data
- ▶ Towards an analytic treatment of understanding the nature of the optimism associated with using the training data to evaluate generalization error, we can consider the *in-sample* error

$$\text{Err} = \frac{1}{n} \sum_{i=1}^n E_y \left[E_{Y^{\text{new}}} \left[L(Y_i^{\text{new}}, \hat{f}(x_i)) \right] \right]$$

- ▶ The notation Y^{new} indicates that we observe n new outcome values *at each of the training points* x_i , $i = 1, \dots, n$

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- ▶ Each of the n components of the in-sample error averages over the randomness in two distributions
 - ▶ the randomness in the observed outcomes in the training data, $\mathbf{y}$
 - ▶ the randomness in the 'new' outcome observation, y_i^{new}
- ▶ The *optimism* is defined as the expected difference between the in-sample error and the training error

$$\text{op} \equiv \text{Err} - E_y [\text{err}]$$

- ▶ expectation is taken with respect to the sampling distribution based on the training data, $\mathbf{y}$

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- For squared error loss, a little algebra leads to

$$\text{op} = \frac{2}{n} \sum_{i=1}^n \text{Cov}[\hat{y}_i, y_i]$$

Assessing EPE

- ▶ This definition leads to the relation

$$\text{Err} = E_y[\text{err}] + \frac{2}{n} \sum_{i=1}^n \text{Cov}[\hat{y}_i, y_i]$$

- ▶ So, the extent to which err is optimistic, as an estimator of Err, depends on how strongly y_i influences its own prediction

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- ▶ The expression simplifies if $\hat{y}_i$ is linear in the y 's

$$\hat{y}_i = \sum_{j=1}^n \pi_j y_j$$

so that

$$\begin{aligned} \text{op} &= \frac{2}{n} \sum_{i=1}^n \mathbb{E}_y [(\hat{y}_i - \mathbb{E}_y[\hat{y}_i])(y_i - \mathbb{E}_y[y_i])] \\ &= \frac{2}{n} \sum_{i=1}^n \pi_i \text{Var}[y_i] \end{aligned}$$

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- ▶ For example, under the additive error model

$$Y = f(X) + \epsilon$$

with $E[\epsilon] = 0$ and $\text{Var}[\epsilon] = \sigma^2$, we obtain

$$\text{Err} = E_y[\text{err}] + \frac{2}{n}p\sigma^2$$

- ▶ p is the number of parameters fit in the regression

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Estimation of in-sample error

- ▶ While decision theory tells us that EPE is a natural criterion for model selection, the in-sample error can still be useful
 - ▶ having an analytic treatment makes the approach convenient
 - ▶ can be effective if we focus on relative differences in error between model options, rather than the absolute error itself
- ▶ From the previous relation, the general form of an estimator for Err is

$$\widehat{\text{Err}} = \text{err} + \widehat{\text{op}}$$

where $\widehat{\text{op}}$ is an estimate of the optimism

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Estimation of in-sample error

Mallow's C_p

- ▶ For the linear model, squared error loss leads to Mallow's C_p statistic:

$$\begin{aligned} C_p &= \text{err} + \frac{2}{n}p\sigma^2 \\ &= \frac{1}{n} \{ \text{RSS} + 2p\hat{\sigma}^2 \} \end{aligned}$$

- ▶ The estimate $\hat{\sigma}^2$ is typically taken from a low-bias model
 - ▶ the most complex model under consideration
- ▶ The C_p statistic penalizes the residual sum of squares by a factor proportional to the number of parameters being estimated
 - ▶ the more complex the model, the greater p will be and the greater the penalty

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Akaike information criterion; AIC

- ▶ The Akaike information criterion is a more general estimate of Err when a log-likelihood function is used as the loss function
 - ▶ for a model parameterized by θ , we take

$$L(Y, f_{\theta}(X)) = -2 \log \Pr_{\theta}(Y | X)$$

- ▶ sometimes referred to as *cross-entropy* loss or *deviance*
 - ▶ multiplying by -2 and taking the log makes the loss for the Normal distribution match the squared error loss
- ▶ We use this loss function all the time as a means for choosing the 'best' model from our training data
 - ▶ minimizing the observed loss is maximum likelihood estimation

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Akaike information criterion; AIC

- ▶ AIC relies on the following relationship

$$-2E_Y [\log \Pr_{\hat{\theta}}(Y|X)] \approx -\frac{2}{n}E_Y[\text{loglike}] + 2\frac{p}{n}$$

- ▶ this relationship holds asymptotically as $n \rightarrow \infty$
- ▶ $\hat{\theta}$ is the maximum likelihood estimate
- ▶ 'loglike' is the maximized log-likelihood

$$\text{loglike} = \sum_{i=1}^n \log \Pr_{\hat{\theta}}(y_i|X)$$

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Akaike information criterion; AIC

- ▶ For any general purpose likelihood AIC is defined as

$$\text{AIC} = -\frac{2}{n}\text{loglike} + 2\frac{p}{n}$$

- ▶ for the Normal model, with $\hat{\sigma}^2$ known, this is equivalent to C_p

Ex: King County birth weight data

Model selection and coefficient shrinkage

- ▶ In many prediction situations there are a large number of inputs, X
- ▶ While it may be the case that $f(X) = X^T \beta$ appropriately describes the underlying mechanisms, it is always the case that we have a finite training sample size, n
- ▶ Prediction accuracy:
 - ▶ least squares estimates may have low bias, but in 'small'-sample settings can exhibit large variability
 - ▶ we could sacrifice a little bias to reduce variation and achieve better overall predictive accuracy

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Ex: King County birth weight data

Model selection and coefficient shrinkage

- ▶ Another issue is interpretation:
 - ▶ with a large number of predictors, it may be hard conceptualize 'holding everything else constant'
 - ▶ may be desirable to restrict attention to a smaller subset of variables which exhibit the strongest effects

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Ex: King County birth weight data

King County birth data

- ▶ As an example, let's consider data on child birth weights for children born in King County, WA in 2001
- ▶ The dataset contains information on a sample of $n=2,500$ births from 2001
- ▶ The data was originally obtained to determine if a new state program ('First Steps') to educate women on proper nutrition during pregnancy was associated with greater birth weight
- ▶ The key outcome variable of interest is birth weight
 - ▶ Birth weight ranges from 255g to 5,175g
 - ▶ 5.1% of babies (127) were born at *low birth weight* ($< 2,500\text{g}$)
- ▶ A total of 15 potential predictor variables are available for investigation

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Ex: King County birth weight data

Complete variable listing

"gender"	M = male, F = female baby
"plural"	1 = singleton, 2 = twin, 3 = triplet
"age"	mother's age in years
"race"	race categories (for mother)
"parity"	number of previous live born infants
"married"	Y = yes, N = no
"bwt"	birth weight in grams
"smokeN"	number of cigarettes smoked per day during pregnancy
"drinkN"	number of alcoholic drinks per week during pregnancy
"firststep"	1 = participant in program; 0 = did not participate
"welfare"	1 = participant in public assistance program; 0 = did not
"smoker"	Y = yes, N = no, U = unknown
"drinker"	Y = yes, N = no, U = unknown
"wpre"	mother's weight in pounds prior to pregnancy
"wgain"	mother's weight gain in pounds during pregnancy
"education"	highest grade completed (add 12 + 1 / year of college)
"gestation"	weeks from last menses to birth of child

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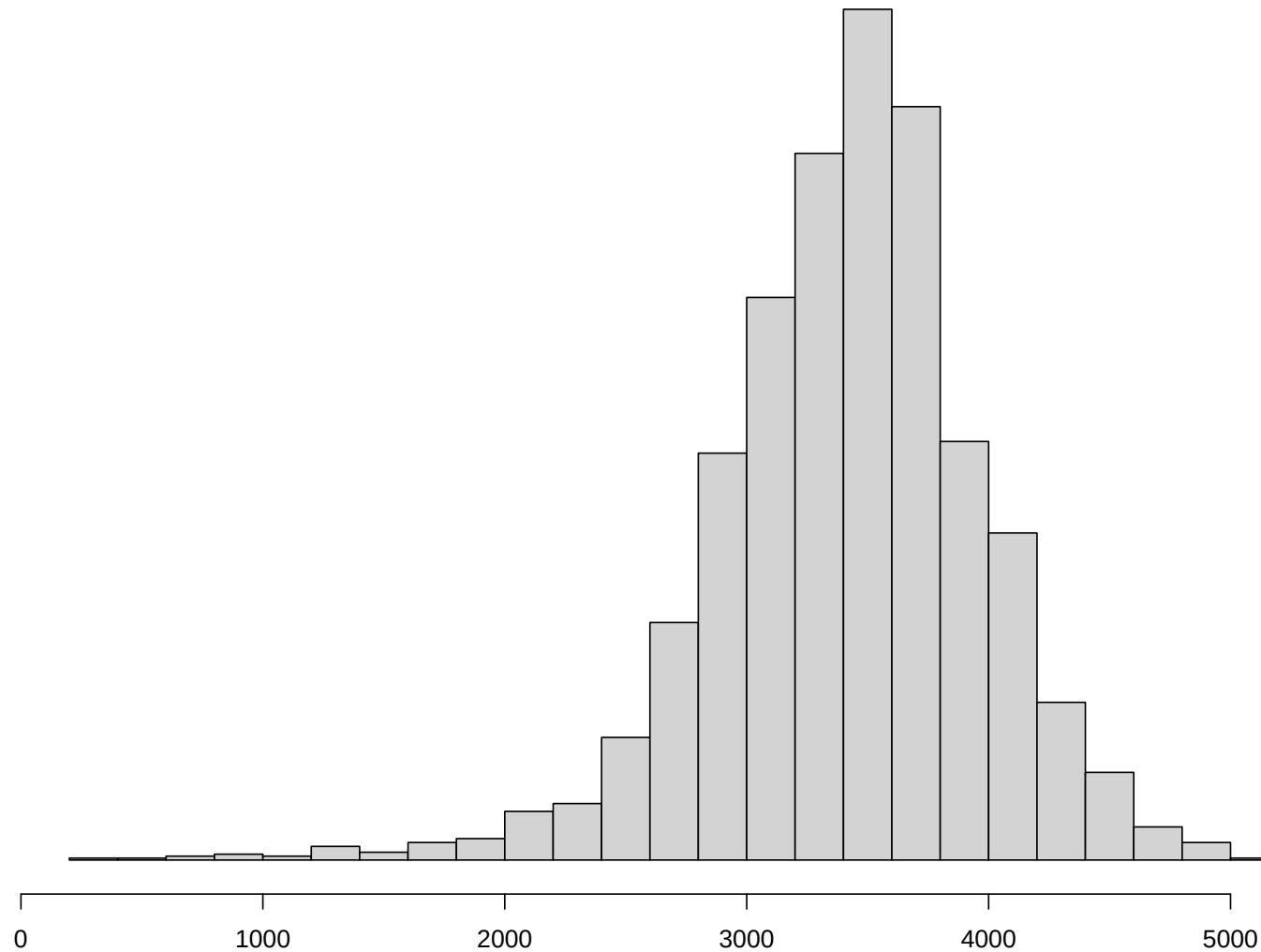
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Distribution of birth weights from the King County data

Birth weight, grams



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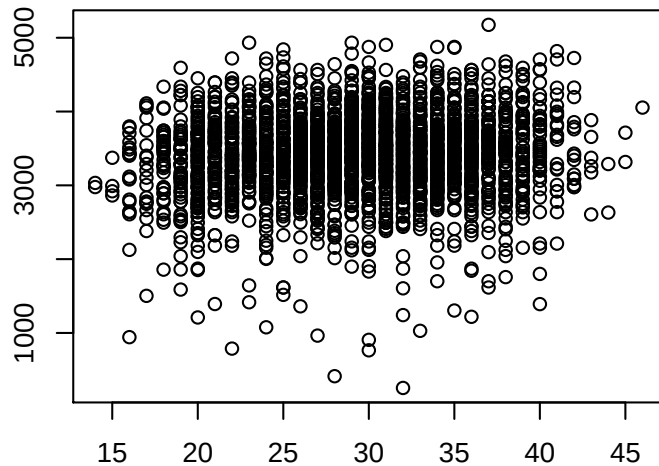
Ex: King County Birth
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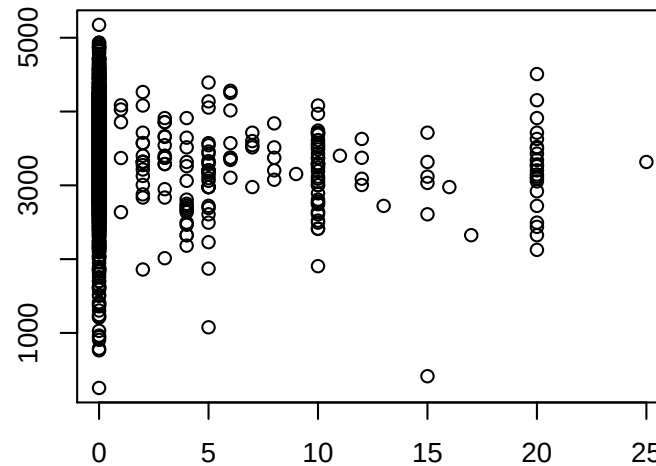
Cross-validation

Selected scatterplots from the King County data

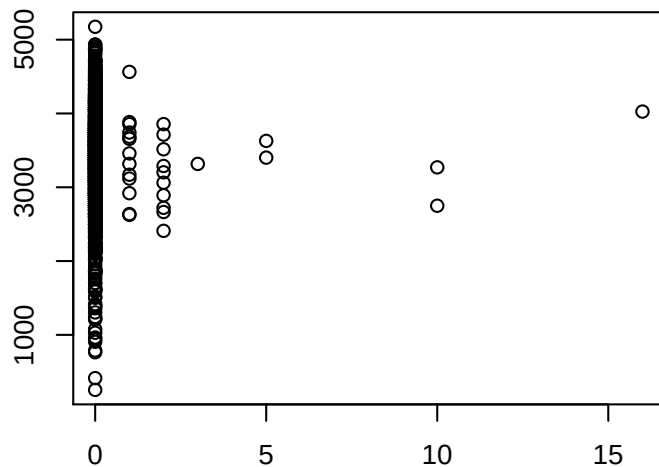
Mother's age, years



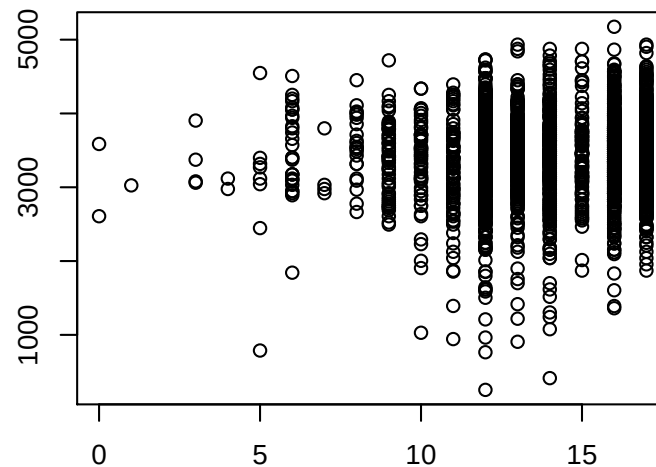
Cigarettes smoked per day



Alcoholic drinks per week



Highest grade completed



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Ex: King County birth weight data

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Subset selection vs. shrinkage

- ▶ Rather than attempting to fit and report a model which includes all the potential predictors, we can consider two strategies
 - ▶ subset selection
 - ▶ shrinkage methods (e.g. LASSO or ridge regression)
- ▶ For now we will focus on subset selection

Ex: King County birth weight data

Subset selection

- ▶ Here we retain only a subset of variables
 - ▶ the remaining variables essentially have their β coefficients set to zero
- ▶ Various strategies exist for ‘choosing’ the variables to keep (or throw out)
 - ▶ best subset selection
 - ▶ stepwise strategies

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Ex: King County birth weight data

Best subsets regression

- ▶ Suppose X consists of p components; $X_1, \dots, X_p$
- ▶ For each $k \in \{1, \dots, p\}$, find the subset of k variables which results in the smallest residual sums of squares
 - ▶ other criteria include Mallows' C_p , R^2 and adjusted R^2
- ▶ Can quickly become computationally intensive when p gets large
- ▶ In R, code is implemented in the `leaps` package

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Ex: King County birth weight data

Best subsets regression in R

```
library(leaps)

## Model with only the intercept
##
fit0 <- lm(bwt ~ 1, data=weight)

## Perform best subsets analysis
##
##   maxModel: a model which includes all the variables you wish to
##             entertain
##   nvmax:    maximum number of variables for the subset selection
##   nbest:    specify, for any given k, the number of the best models
##             are to be returned
##
maxModel <- as.formula(bwt ~ gender + age + race + parity + married
                      + smokeN + drinkN + firststep + welfare
                      + smoker + drinker + wpre + education )

bestSub  <- summary(regsubsets(maxModel, data=weight,
                              nvmax=17, nbest=10))
```

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Ex: King County birth weight data

Best subsets regression in R

```
names(bestSub)
[1] "which" "rsq"    "rss"    "adjr2" "cp"    "bic"
     "outmat" "obj"

## 'results' contains the subset size, k, and the residual
## sum of squares

results <- c(0, sum((weight$bwt- fitted(fit0))^2))
results <- rbind(results,
                 cbind(apply(bestSub$which, 1, sum)-1, bestSub$rss))

##
##### Look at minimum residual sums of squares
##
minRSS <- tapply(results[,2], results[,1], FUN=min)
```

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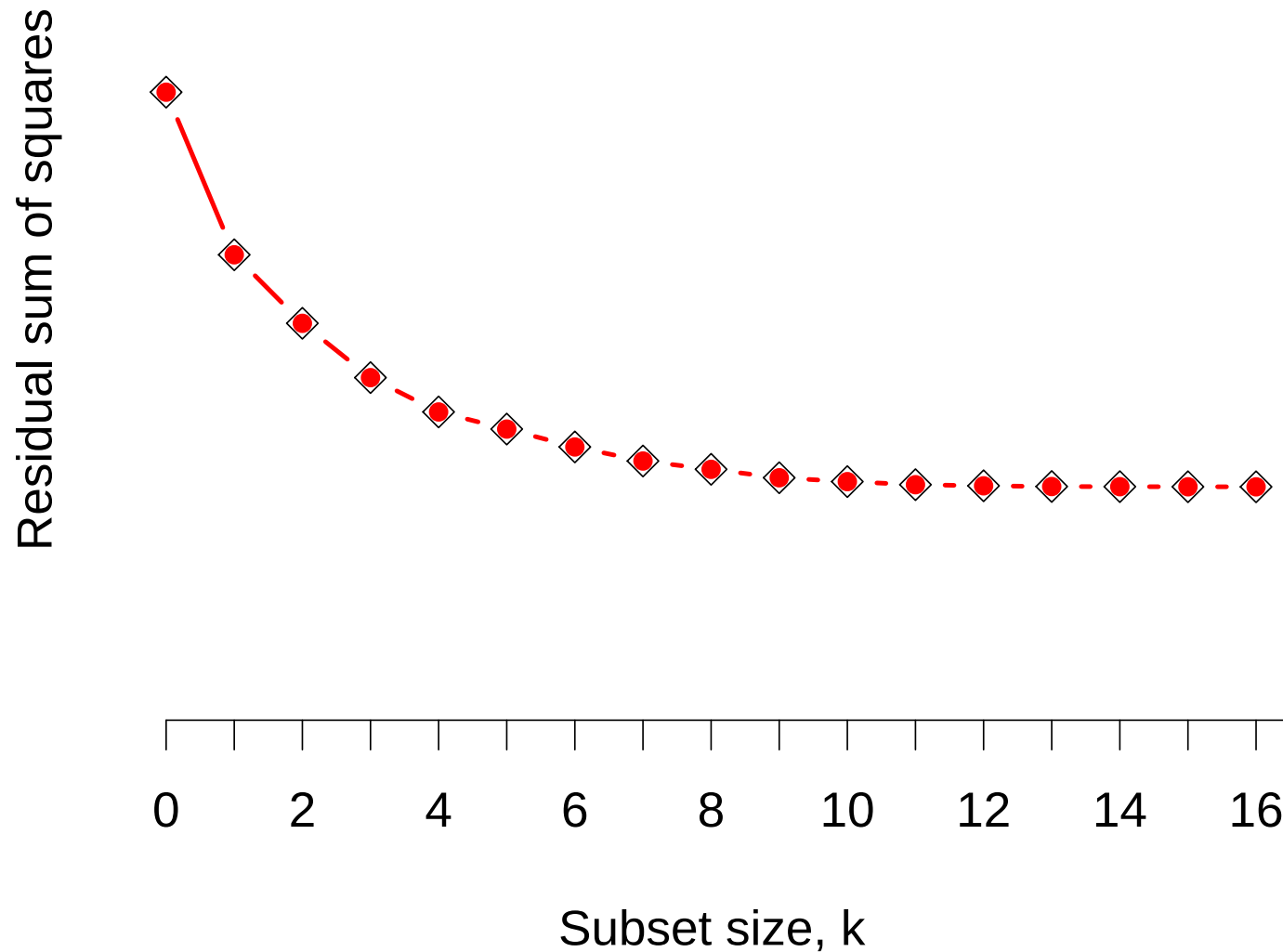
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Best subset selection for the King County 2001 birth weight data



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Ex: King County birth weight data

Best subsets regression

- ▶ The best-subset curve is necessarily decreasing, so it cannot be used as a criteria for choosing k
- ▶ Typically choose a model which minimizes an estimate of the EPE
 - ▶ Mallow's C_p , AIC, BIC, cross-validation

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Best subsets regression in R

```
##
#####
#####      Now let's do best subsets with Cp as the criteria
#####
##
bestSubCp  <- leaps(x=model.matrix(fitF),
                    y=weight$bwt, int=FALSE,
                    nbest=1, method="Cp")

## 'results' contains the subset size, k, and the Cp value
##
results <- NULL
results <- rbind(results, cbind(apply(bestSubCp$which, 1, sum)-1,
                                   bestSubCp$Cp))

##
minCp<- tapply(results[,2], results[,1], FUN=min)
```

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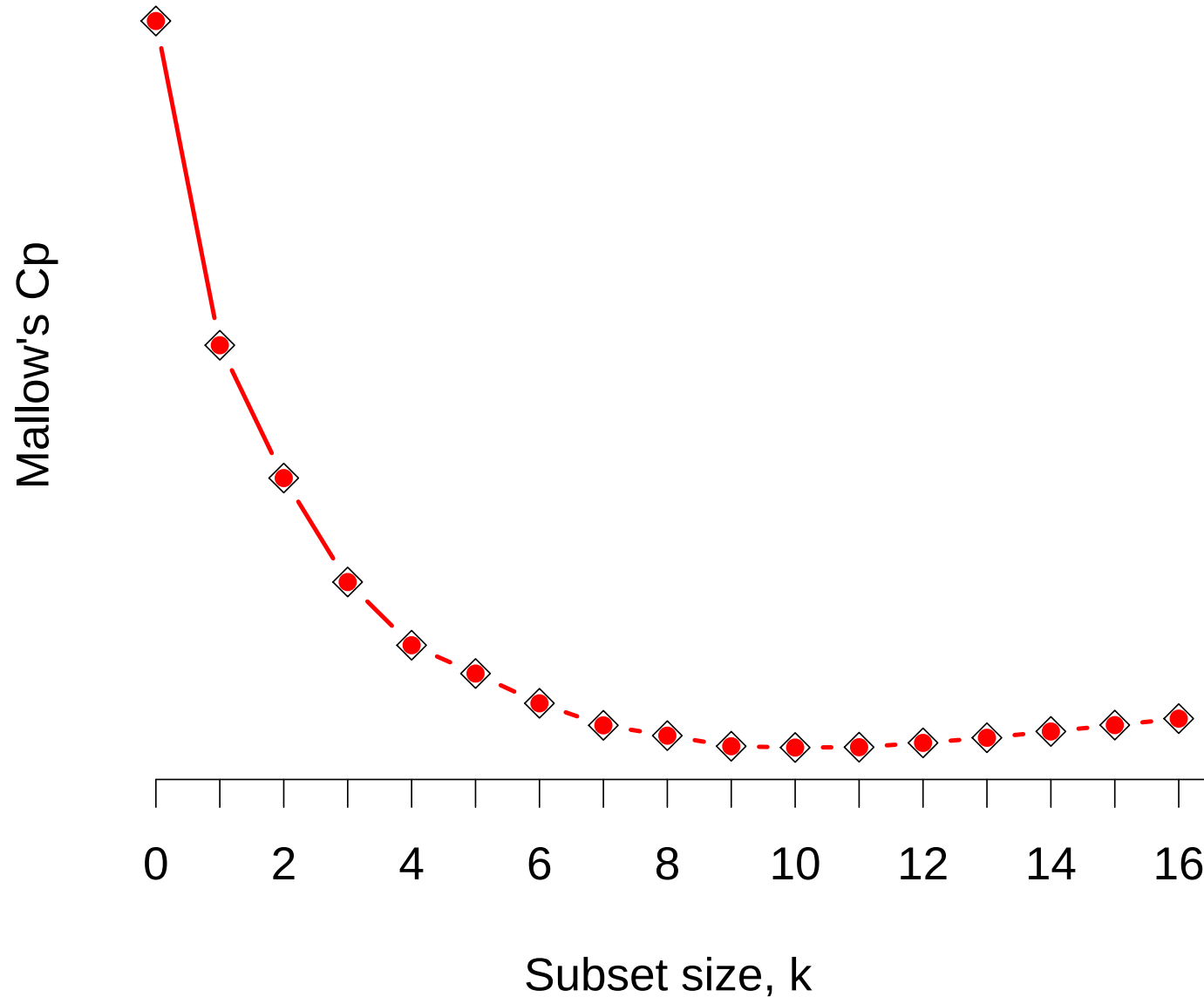
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Best subsets regression in R

```
> ##
> #####
> #####           Which covariates were selected in the k=10 model...
> #####
> ##
> cbind( dimnames( model.matrix(fitF) )[[2]],bestSubCp$which[10,])
  [,1]      [,2]
1 "(Intercept)" "TRUE"
2 "genderM"      "TRUE"
3 "age"          "FALSE"
4 "raceblack"    "FALSE"
5 "racehispanic" "TRUE"
6 "raceother"    "FALSE"
7 "racewhite"    "TRUE"
8 "parity"       "TRUE"
9 "marriedYes"   "TRUE"
A "smokeN"       "TRUE"
B "drinkN"       "FALSE"
C "firststepYes" "FALSE"
D "welfareYes"   "TRUE"
E "smokerY"      "FALSE"
F "drinkerY"     "FALSE"
G "wpre"         "TRUE"
H "educ"         "TRUE"
```

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Ex: King County birth weight data

Stepwise procedures

- ▶ Instead of performing an exhaustive enumeration for each value for k , we can search for a 'good path'
- ▶ Forward selection
 - ▶ start with an 'intercept-only' model and build up the model
- ▶ Backward selection
 - ▶ start with a 'full' model and reduce the model
- ▶ In R, see `stepAIC` in the `MASS` library

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Model Assessment
and Selection

Assessing EPE

Estimation of
In-Sample Error

Ex: King County Birth
Weight Data

Best subsets regression

Cross-validation

Ex: King County birth weight data

Stepwise AIC in R

```
library(MASS)
##
#####
#####      Stepwise selection using AIC
#####
##
fitStepAIC <- stepAIC( fit0, scope=maxModel, direction="forward" )
.
.
.
>Step:  AIC=31425
bwt ~ wpre + smoker + gender + married + race + parity + welfare +
      education + smokeN
```

	Df	Sum of Sq	RSS	AIC
<none>			7.12e+08	3.14e+04
+ drinker	1	1.96e+05	7.12e+08	3.14e+04
+ drinkN	1	4.37e+04	7.12e+08	3.14e+04
+ firststep	1	2.41e+04	7.12e+08	3.14e+04
+ age	1	1.88e+04	7.12e+08	3.14e+04

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K-fold cross-validation

- ▶ Cross-validation involves splitting the data into multiple partitions, training a model on one partition and then applying it to the holdout partition
- ▶ K-fold cross-validation involves splitting the data into $K < n$ parts as follows

(1) Split the data into K roughly equal parts, and denote the collection of indexes for the k^{th} part as C_k , $k = 1, \dots, K$

(2) For each part, fit a model using all the remaining data,

$$\mathbf{y}^{(k)} = \{y_i \mid i \notin C_k\},$$

and denote the fitted model as $\hat{f}^k(x)$

(3) For all i such that $i \in C_k$, obtain a prediction via

$$\hat{y}_i = \hat{f}^k(x_i)$$

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- ▶ Let $k(i)$ denote the part in which y_i resides
- ▶ The K -fold cross validation statistic, for a general loss function, is given by

$$CV_K = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}^{-k(i)}(x_i))$$

where L denotes a *loss function*. (We have been considering squared error loss so that

$$L(y_i, \hat{f}^{-k(i)}(x_i)) = (y_i - \hat{f}^{-k(i)}(x_i))^2$$

- ▶ As we decrease K , however, the bias of CV_K increases
 - ▶ CV_K is biased upward
 - ▶ extent depends on the sample size

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